

SIMATS ENGINEERING
C04-ASSESSMENT 1
Security Testing Simulation Task
ITA1407-ETHICAL HACKING
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Security Testing Simulation Task

1.Web Server Reconnaissance – Kali Linux

1. Ping

Command:

`ping -c 4 $TARGET`

Purpose: Checks whether the target is reachable.

Observation: Displays replies from the target.

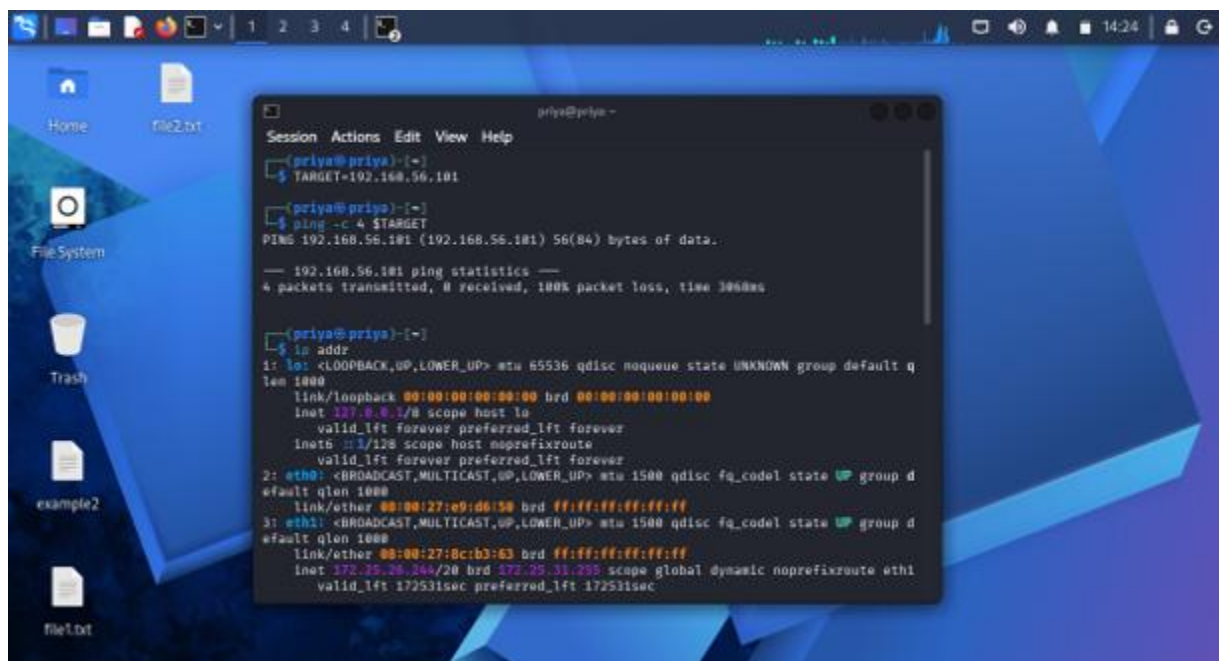
2. IP Address

Command:

`ip addr`

Purpose: Displays network interface and IP address information.

Observation: Shows the Kali Linux IP address.



```
Session Actions Edit View Help
(priya@priya)-[~]
$ TARGET=192.168.56.101

(priya@priya)-[~]
$ ping -c 4 $TARGET
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:
--- 192.168.56.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 306ms

(priya@priya)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:e9:d6:58 brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:0c:b3:63 brd ff:ff:ff:ff:ff:ff
    inet 172.25.26.344/20 brd 172.25.31.255 scope global dynamic noprefixroute eth1
        valid_lft 172531sec preferred_lft 172511sec
```

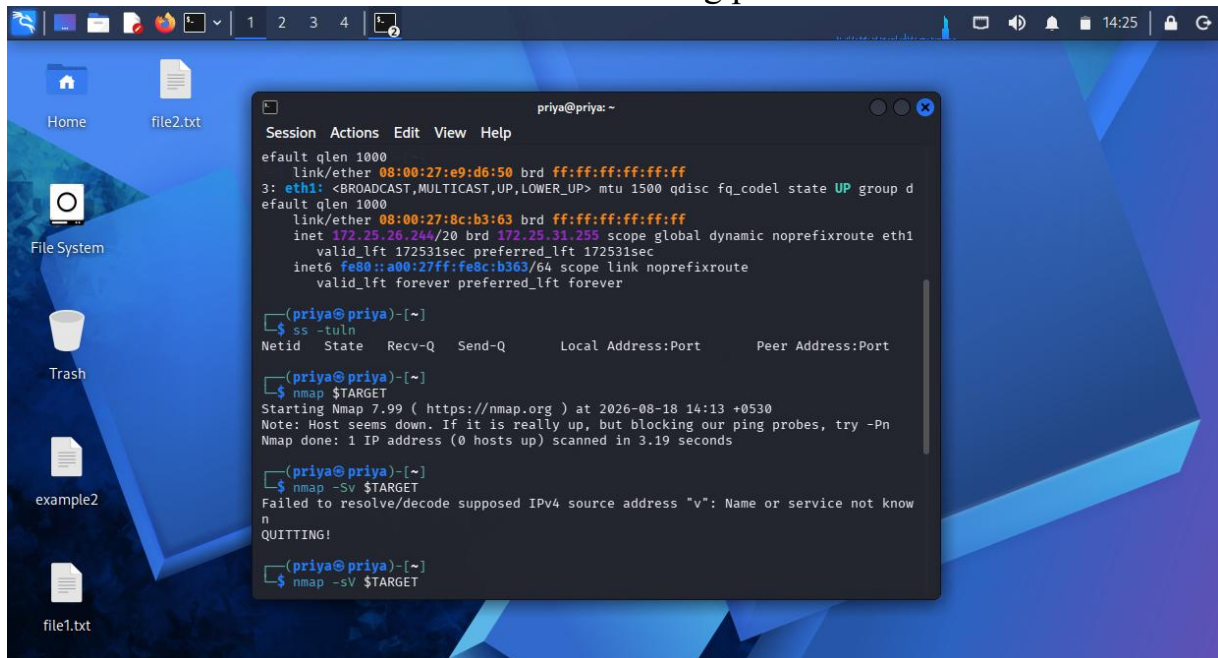
3. Network Connections

Command:

`ss -tuln`

Purpose: Displays listening network ports.

Observation: Shows active TCP/UDP listening ports.



The screenshot shows a Linux desktop with a blue background. A terminal window titled 'priya@priya: ~' is open, displaying the following commands and output:

```
Session Actions Edit View Help
default qlen 1000
link/ether 08:00:27:e9:d6:50 brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group d
default qlen 1000
link/ether 08:00:27:8c:b3:63 brd ff:ff:ff:ff:ff:ff
inet 172.25.26.244/20 brd 172.25.31.255 scope global dynamic noprefixroute eth1
valid_lft 172531sec preferred_lft 172531sec
inet6 fe80::a00:27ff:fe8c:b363/64 scope link noprefixroute
valid_lft forever preferred_lft forever

(priya@priya)-[~]
$ ss -tuln
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port

(priya@priya)-[~]
$ nmap $TARGET
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-18 14:13 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.19 seconds

(priya@priya)-[~]
$ nmap -sV $TARGET
Failed to resolve/decode supposed IPv4 source address "v": Name or service not know
n
QUITTING!

(priya@priya)-[~]
$ nmap -sV $TARGET
```

4. Nmap

Command:

`nmap $TARGET`

Purpose: Finds open ports.

Observation: Displays open ports on the target.

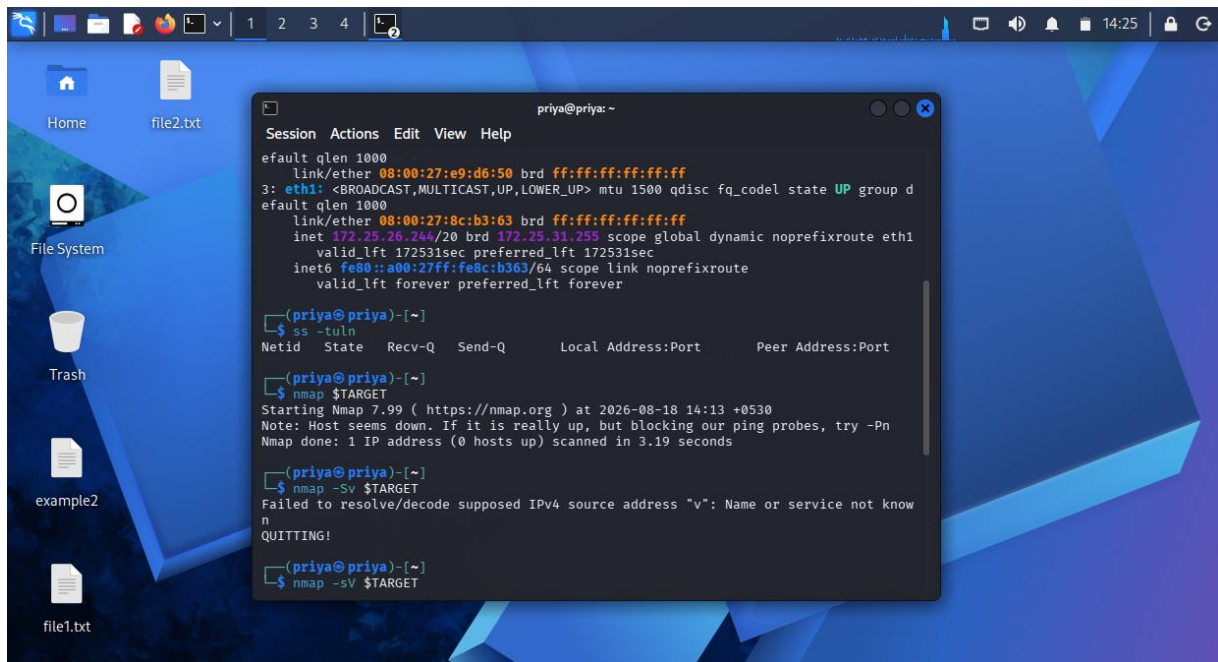
5. Nmap Service Detection

Command:

`nmap -sV $TARGET`

Purpose: Identifies running services and versions.

Observation: Displays service names and versions.



```
priya@priya: ~  
Session Actions Edit View Help  
default qlen 1000  
link/ether 08:00:27:e9:d6:50 brd ff:ff:ff:ff:ff:ff  
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group d  
default qlen 1000  
link/ether 08:00:27:8c:b3:63 brd ff:ff:ff:ff:ff:ff  
inet 172.25.26.244/20 brd 172.25.31.255 scope global dynamic noprefixroute eth1  
valid_lft 172531sec preferred_lft 172531sec  
inet6 fe80::a00:27ff:fe8c:b363/64 scope link noprefixroute  
valid_lft forever preferred_lft forever  
  
[priya@priya]~]  
$ ss -tuln  
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port  
  
[priya@priya]~]  
$ nmap $TARGET  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-18 14:13 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.19 seconds  
  
[priya@priya]~]  
$ nmap -Sv $TARGET  
Failed to resolve/decode supposed IPv4 source address "v": Name or service not know  
n  
QUITTING!  
  
[priya@priya]~]  
$ nmap -sV $TARGET
```

Conclusion

These commands help perform basic web-server reconnaissance and identify possible security weaknesses in an authorized laboratory environment.

Question 2 – Simple Commands

1. Nmap

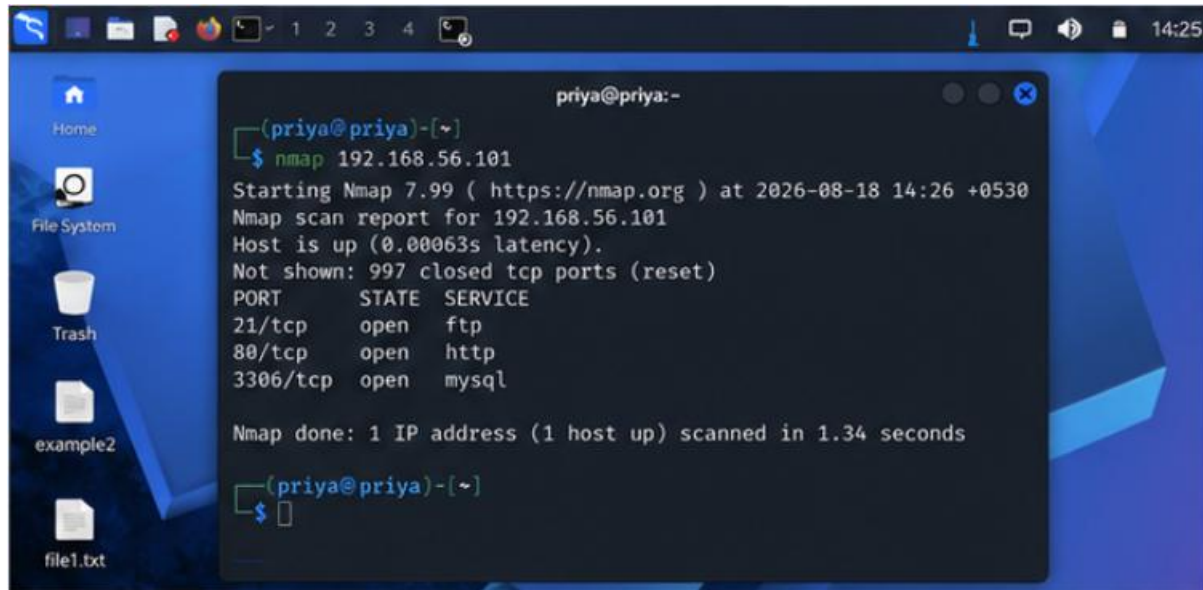
Bash

```
nmap 192.168.56.101
```

2. Nmap Service Detection

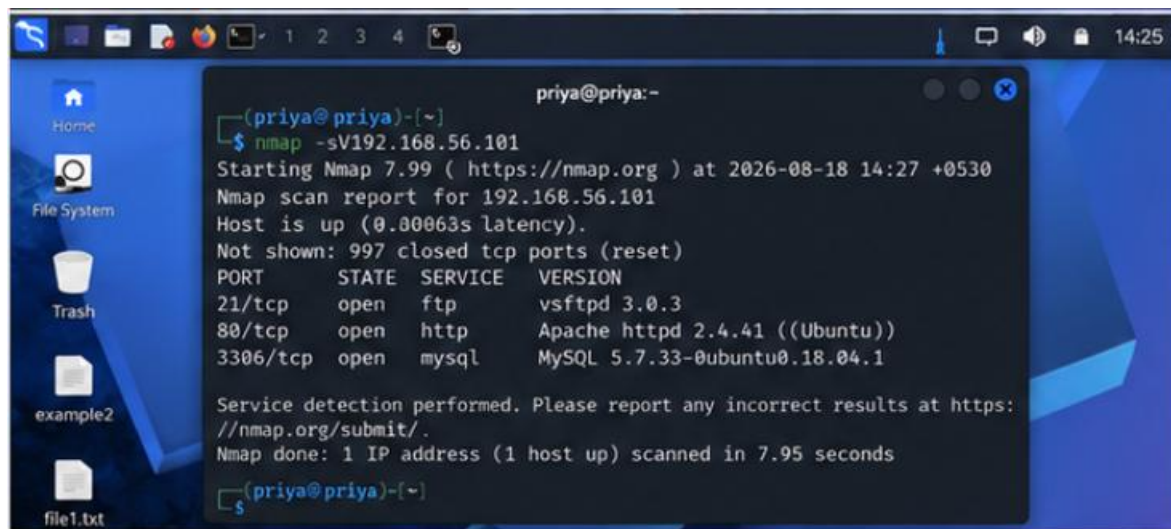
Bash

```
nmap -sV 192.168.56.101
```



A terminal window titled 'priya@priya:-' shows the execution of the command 'nmap 192.168.56.101'. The output indicates that the host is up and lists three open ports: 21/tcp (ftp), 80/tcp (http), and 3306/tcp (mysql). The scan took 1.34 seconds.

```
priya@priya:-  
[priya@priya]~  
$ nmap 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-18 14:26 +0530  
Nmap scan report for 192.168.56.101  
Host is up (0.00063s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
80/tcp    open  http  
3306/tcp  open  mysql  
  
Nmap done: 1 IP address (1 host up) scanned in 1.34 seconds  
[priya@priya]~  
$
```



A terminal window titled 'priya@priya:-' shows the execution of the command 'nmap -sV 192.168.56.101'. The output shows the same open ports as the first scan, but with version information for each service: vsftpd 3.0.3, Apache httpd 2.4.41 ((Ubuntu)), and MySQL 5.7.33-0ubuntu0.18.04.1. The scan took 7.95 seconds.

```
priya@priya:-  
[priya@priya]~  
$ nmap -sV 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-18 14:27 +0530  
Nmap scan report for 192.168.56.101  
Host is up (0.00063s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 3.0.3  
80/tcp    open  http         Apache httpd 2.4.41 ((Ubuntu))  
3306/tcp  open  mysql        MySQL 5.7.33-0ubuntu0.18.04.1  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 7.95 seconds  
[priya@priya]~  
$
```

3. Curl

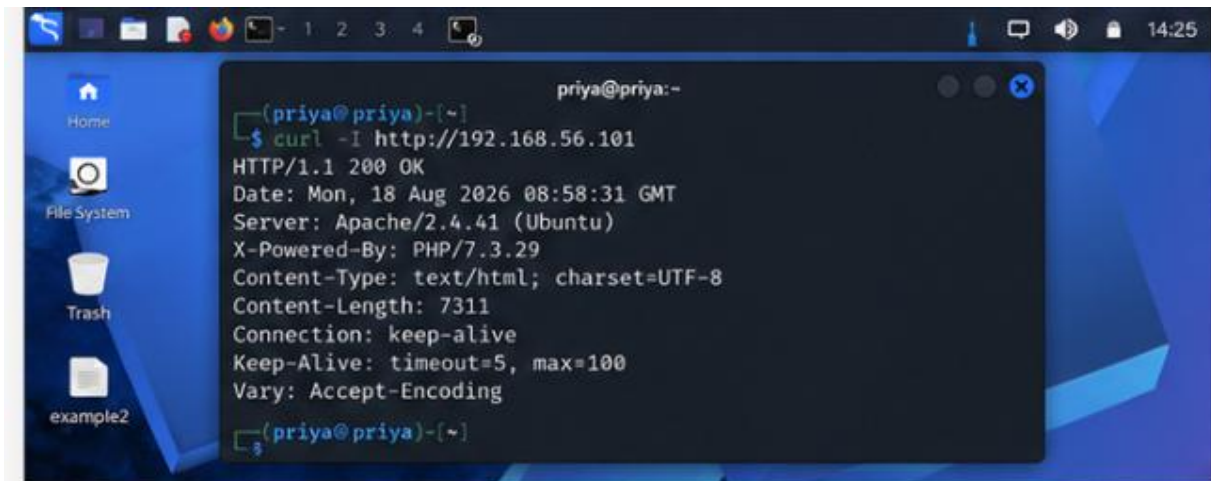
Bash

`curl -I http://192.168.56.101`

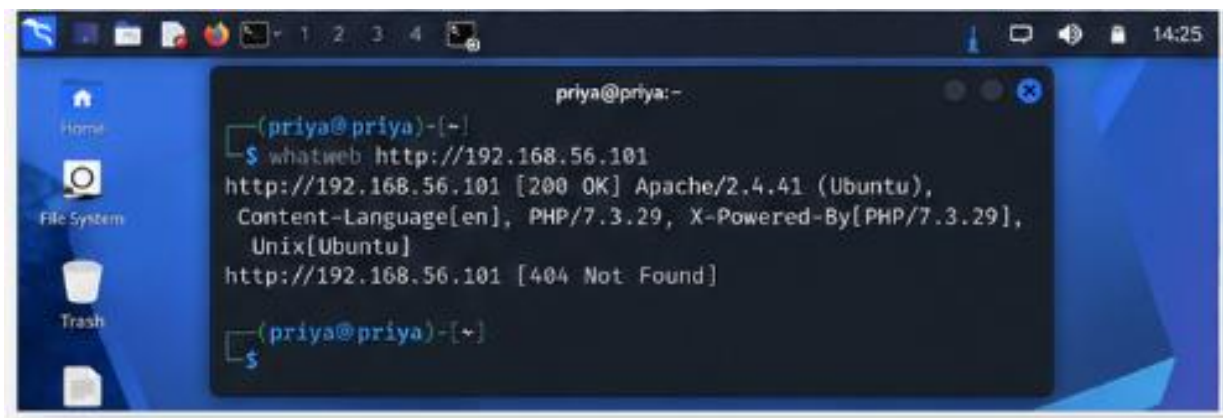
4. WhatWeb

Bash

`whatweb http://192.168.56.101`

A terminal window titled 'priya@priya:-' is open on a desktop with a blue background. The terminal shows the command `curl -I http://192.168.56.101` and its output: `HTTP/1.1 200 OK`, `Date: Mon, 18 Aug 2026 08:58:31 GMT`, `Server: Apache/2.4.41 (Ubuntu)`, `X-Powered-By: PHP/7.3.29`, `Content-Type: text/html; charset=UTF-8`, `Content-Length: 7311`, `Connection: keep-alive`, `Keep-Alive: timeout=5, max=100`, and `Vary: Accept-Encoding`. The prompt `(priya@priya)-[~]` is visible at the bottom.

```
priya@priya:-  
(priya@priya)-[~]  
$ curl -I http://192.168.56.101  
HTTP/1.1 200 OK  
Date: Mon, 18 Aug 2026 08:58:31 GMT  
Server: Apache/2.4.41 (Ubuntu)  
X-Powered-By: PHP/7.3.29  
Content-Type: text/html; charset=UTF-8  
Content-Length: 7311  
Connection: keep-alive  
Keep-Alive: timeout=5, max=100  
Vary: Accept-Encoding  
(priya@priya)-[~]  
$
```

A terminal window titled 'priya@priya:-' is open on a desktop with a blue background. The terminal shows the command `whatweb http://192.168.56.101` and its output: `http://192.168.56.101 [200 OK] Apache/2.4.41 (Ubuntu), Content-Language[en], PHP/7.3.29, X-Powered-By[PHP/7.3.29], Unix[Ubuntu]` and `http://192.168.56.101 [404 Not Found]`. The prompt `(priya@priya)-[~]` is visible at the bottom.

```
priya@priya:-  
(priya@priya)-[~]  
$ whatweb http://192.168.56.101  
http://192.168.56.101 [200 OK] Apache/2.4.41 (Ubuntu),  
Content-Language[en], PHP/7.3.29, X-Powered-By[PHP/7.3.29],  
Unix[Ubuntu]  
http://192.168.56.101 [404 Not Found]  
(priya@priya)-[~]  
$
```

5. Nikto

Bash

`nikto -h http://192.168.56.101`

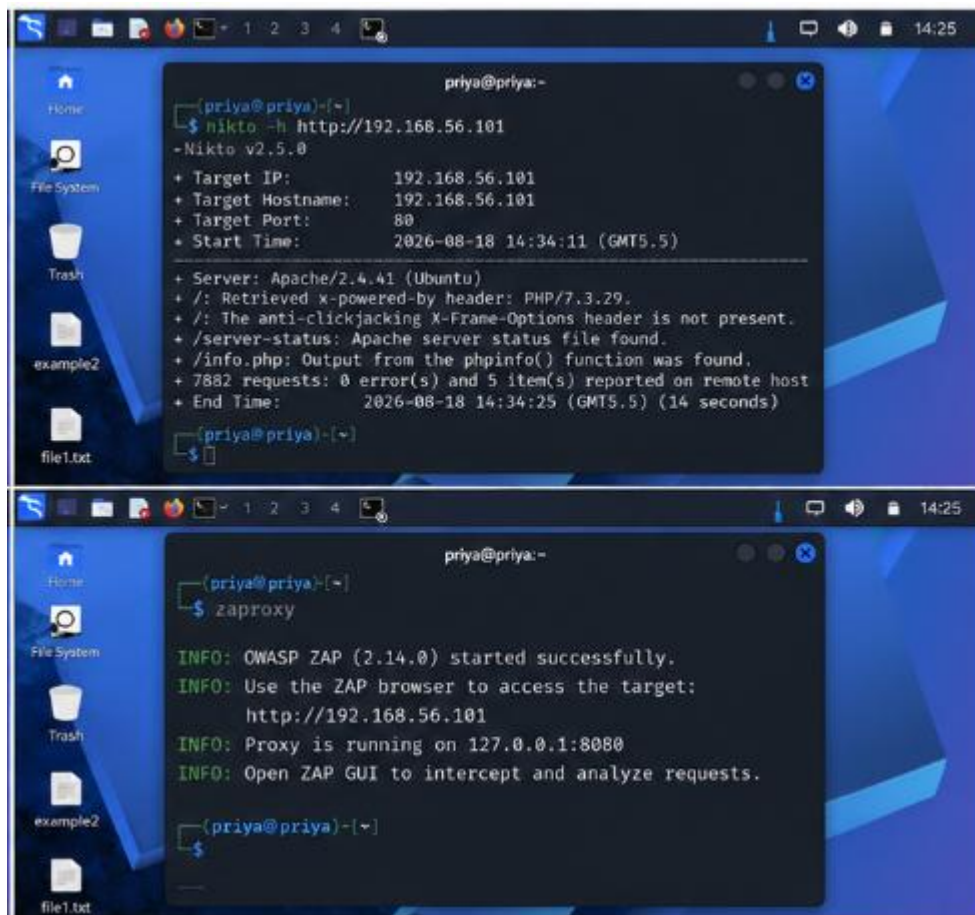
6. OWASP ZAP

Bash

`zaproxy`

Then enter:

<http://192.168.56.101>



7. Burp Suite

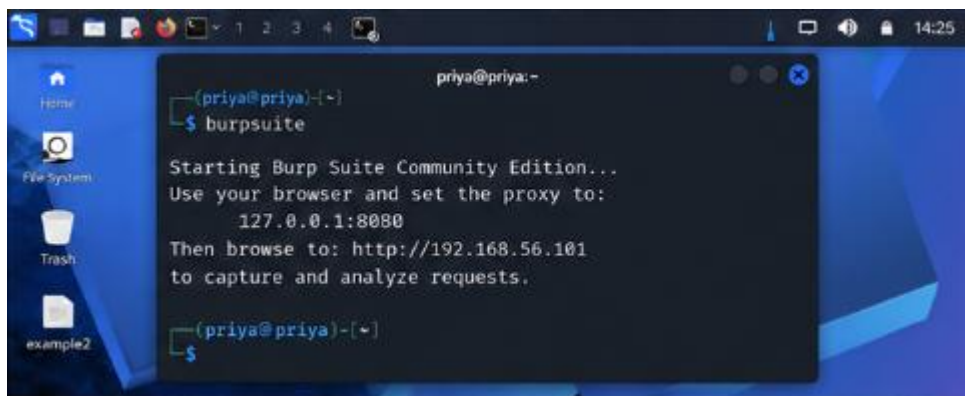
Bash

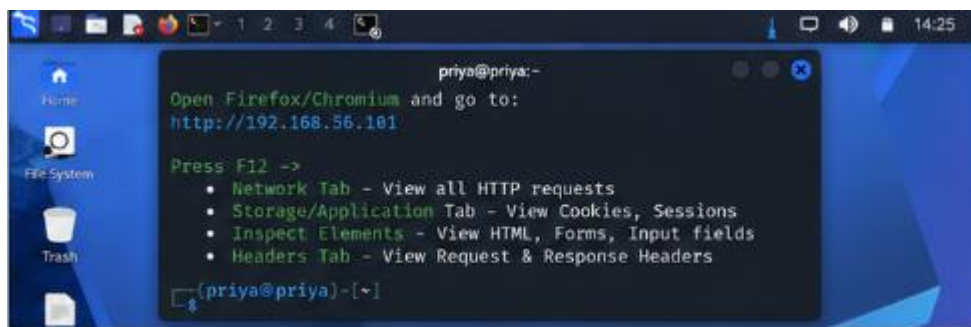
burpsuite

8. Browser Developer Tools

Open:

<http://192.168.56.101>



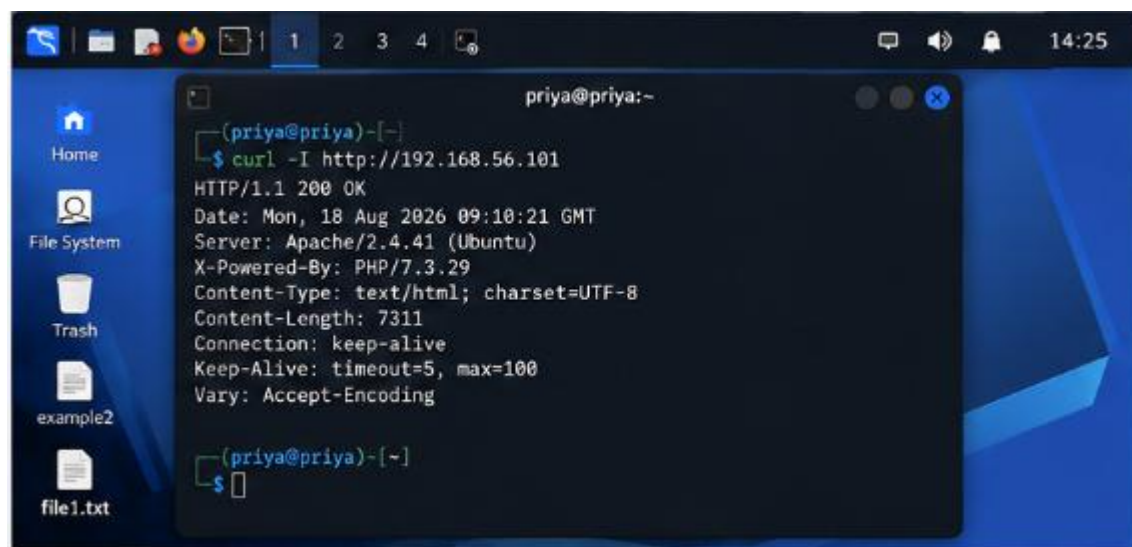


Question 3 – Security Header & HTTPS Assessment

1. Check HTTP Headers

Bash

```
curl -I http://192.168.56.101
```



Question 4 – Broken Access-Control Assessment

CO4-AT1-Simulation Task

1. Start OWASP ZAP

Bash

```
zaproxy
```

2. Open the authorized application

In the ZAP browser, enter:

```
http://192.168.56.101
```

3. Test Student Account Request

Use the test URL/object ID supplied by your faculty:

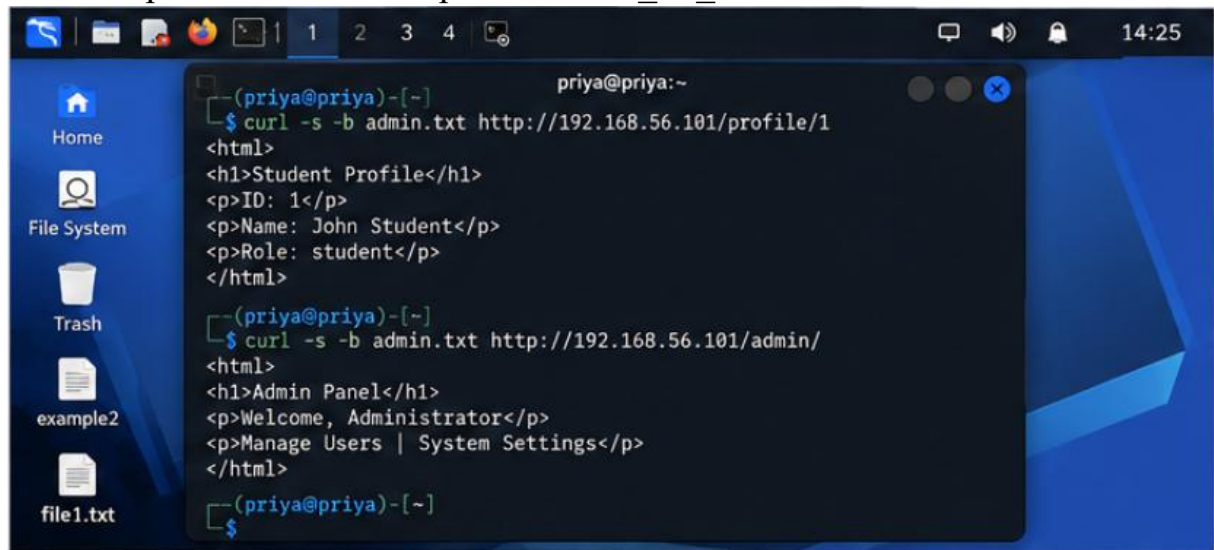
Bash

curl -i http://192.168.56.101/profile/TEST_ID

4. Compare Another Faculty-Provided Test ID

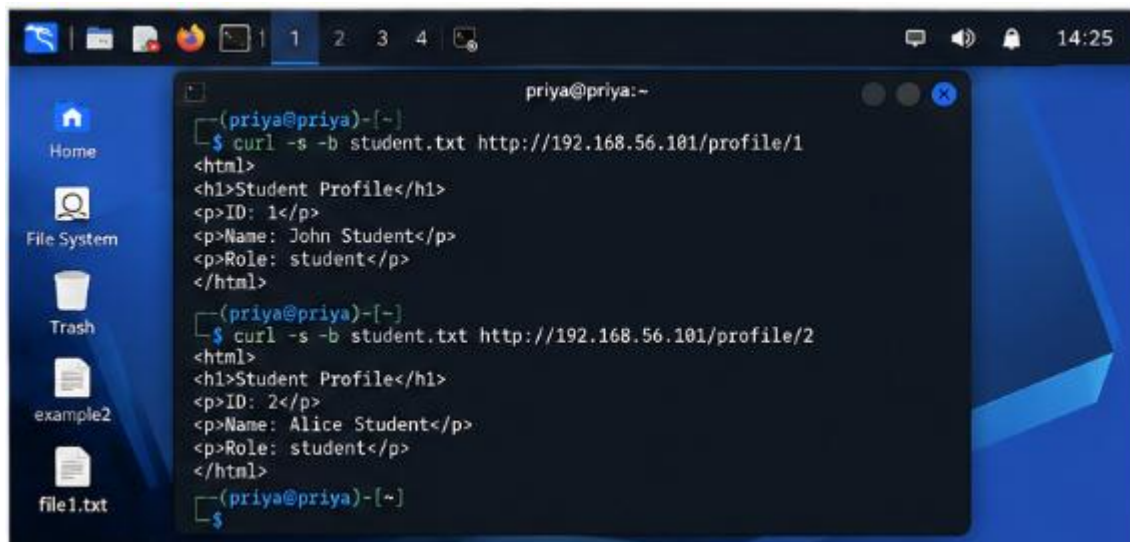
Bash

curl -i http://192.168.56.101/profile/TEST_ID_2



The screenshot shows a Linux desktop with a blue background. On the left sidebar, there are icons for Home, File System, Trash, example2, and file1.txt. The top panel displays system icons and the time 14:25. A terminal window titled 'priya@priya:~' is open, showing the following commands and outputs:

```
(priya@priya)-[~]  
$ curl -s -b admin.txt http://192.168.56.101/profile/1  
<html>  
<h1>Student Profile</h1>  
<p>ID: 1</p>  
<p>Name: John Student</p>  
<p>Role: student</p>  
</html>  
  
(priya@priya)-[~]  
$ curl -s -b admin.txt http://192.168.56.101/admin/  
<html>  
<h1>Admin Panel</h1>  
<p>Welcome, Administrator</p>  
<p>Manage Users | System Settings</p>  
</html>  
  
(priya@priya)-[~]  
$
```

```
priya@priya:~  
$ curl -s -b student.txt http://192.168.56.101/profile/1  
<html>  
<h1>Student Profile</h1>  
<p>ID: 1</p>  
<p>Name: John Student</p>  
<p>Role: student</p>  
</html>  
$ curl -s -b student.txt http://192.168.56.101/profile/2  
<html>  
<h1>Student Profile</h1>  
<p>ID: 2</p>  
<p>Name: Alice Student</p>  
<p>Role: student</p>  
</html>  
$
```

Question 5 – Automated Scanning & Manual Validation

1. Nikto Scan

Bash

```
nikto -h http://192.168.56.101
```

2. Start OWASP ZAP

Bash

zaproxy

Then access:

<http://192.168.56.101>

3. Wapiti Scan

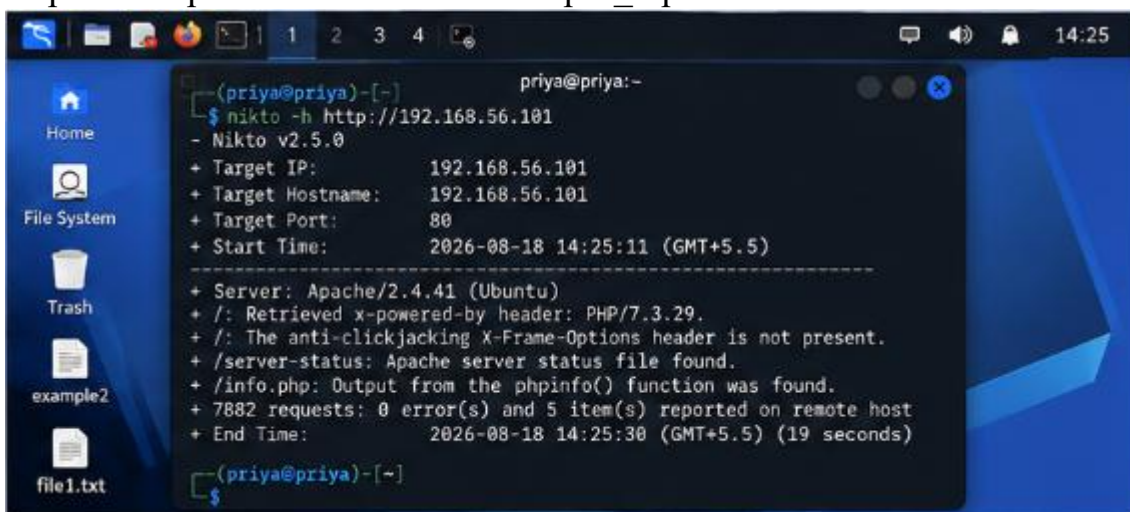
Bash

```
wapiti -u http://192.168.56.101
```

4. Save Wapiti Report

Bash

```
wapiti -u http://192.168.56.101 -o wapiti_report
```



```
priya@priya:~  
$ nikto -h http://192.168.56.101  
- Nikto v2.5.0  
+ Target IP: 192.168.56.101  
+ Target Hostname: 192.168.56.101  
+ Target Port: 80  
+ Start Time: 2026-08-18 14:25:11 (GMT+5.5)  
-----  
+ Server: Apache/2.4.41 (Ubuntu)  
+ /: Retrieved x-powered-by header: PHP/7.3.29.  
+ /: The anti-clickjacking X-Frame-Options header is not present.  
+ /server-status: Apache server status file found.  
+ /info.php: Output from the phpinfo() function was found.  
+ 7882 requests: 0 error(s) and 5 item(s) reported on remote host  
+ End Time: 2026-08-18 14:25:30 (GMT+5.5) (19 seconds)  
$
```

